

PAPER_682: Numerical UQFF Stability Analysis for Sagittarius A*

Author: Daniel Murphy
Framework: Unified Quantum Field Framework (UQFF) v5.30
Session: 173
Date: April 1, 2026
Class: UQFFStabilityNumericallyForSgrA | CP4 #266
Source: grok_share_fc21e30c24b4.txt
Domain: Black Hole Stability / UQFF

Abstract

We perform a four-pronged numerical stability analysis of the UQFF solution for Sgr A* ($M = 4.297 \times 10^6 M_\odot$): (1) perturbation expansion with imaginary frequency $\omega_I^{\text{UQFF}} < 0$ (damped), (2) Lyapunov exponent $\lambda_{\text{UQFF}} = -(\rho_{\text{SCm}}/\rho_{\text{UA}})e^{-U_m/k_B T_H}/\tau_{\text{std}} < 0$, (3) RK4 mass evolution $M(t)$ confirming quasi-static behaviour over 10^{60} s, (4) fixed-point stability classification.

Primary UQFF Equation

$$\lambda_{\text{UQFF}} = -\frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \frac{e^{-U_m/k_B T_H}}{\tau_{\text{std}}(M)} < 0$$

Parameters

Parameter	Value	Description
$M_{\text{Sgr A*}}$	$4.297 \times 10^6 M_\odot$	Black hole mass
T_H	1.4×10^{-14} K	Hawking temperature
λ	< 0	Lyapunov exponent

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m ³	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m ³	Schwarzschild condensate

Constant	Value	Description
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFStabilityNumericallyForSgrA.h"
```

```
UQFFStabilityNumericallyForSgrA obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFStabilityNumericallyForSgrACalculator

calc = UQFFStabilityNumericallyForSgrACalculator()
result = calc.compute() # returns all UQFF quantities
print(result)
```

References

Event Horizon Telescope 2022 (Sgr A*); UQFF Session 173

UQFF v5.30 | Session 173 | April 1, 2026 | PAPER_682 of 1000